

Time spent: 0 sec

Subject:
Biochemistry

QID:403742

System:
Amino Acids and Proteins

The visual phototransduction pathway in mammals is the process by which visual stimulation in the form of photons of light is biochemically received and converted into electrical signals that can be processed by the brain to form images. This phototransduction pathway begins when photons strike rod and cone photoreceptor cells located at the back of the retina. Instead of dendrites, each photoreceptor nerve cell contains hundreds of lipid membrane discs, each of which is filled with opsins, a type of G protein–coupled receptor.

In place of a conventional agonist, opsins are instead covalently bound to a prosthetic group known as retinal. Retinal is attached to the opsin protein through the formation of a Schiff base between the electrophilic aldehyde group of retinal and a nucleophilic primary amine of an amino acid side chain. In the dark, the retinal prosthetic group exists in the 11-*cis* conformation, but energy input from photons causes isomerization to the all-*trans* conformation. In the presence of the light-stimulated all-*trans* isomer, the opsins activate their associated heterotrimeric G protein, known as transducin.

Once activated, transducin then activates cGMP phosphodiesterase, which catalyzes the degradation of 3',5'-cyclic GMP (cGMP). Cyclic nucleotide–gated (CNG) channels are Na⁺- and Ca²⁺-permeable channels that open upon binding to cGMP, allowing extracellular cations to enter. With the degradation of cGMP, these CNG channels close, causing gradual changes in the photoreceptor membrane potential and gradual changes in neurotransmitter release. In the retina, the inputs from the various rod and cone cells are processed by various interneurons—bipolar cells, horizontal cells, and amacrine cells—each with their own graded membrane potential, before reaching the ganglion cells at the front side of the retina. The ganglion cells then process all excitatory and inhibitory inputs in an all-or-none fashion to either produce or not produce an action potential that then travels to the brain for further cognitive processing.

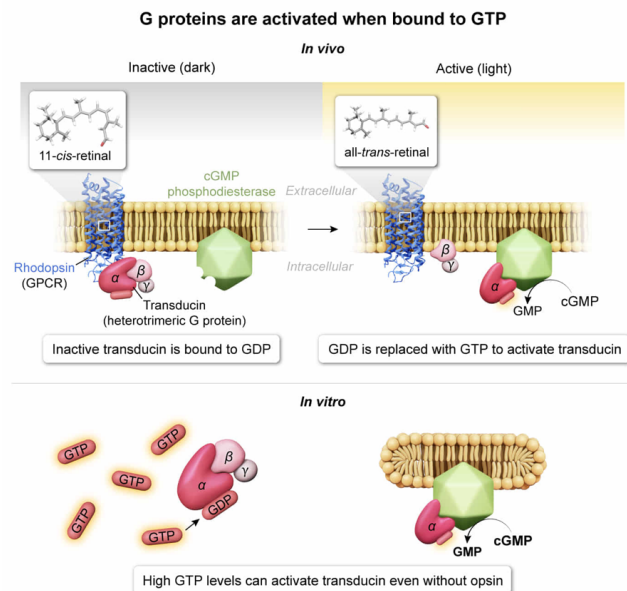
A scientist purified the transducin protein and all downstream components for further study *in vitro*. Addition of which of the following will be necessary to activate the purified transducin?

- ☐ A. Electric stimulation [5%]
- ✓ ☒ B. High levels of GTP [33%]
- ☐ C. Irradiation by light energy [55%]
- ☐ D. Addition of nonspecific protein kinases [5%]

Correct

33% answered correctly

Explanation:



G protein–coupled receptors (GPCRs) are 7-transmembrane **integral membrane proteins** that serve a critical role in many **cell signaling pathways**. In the resting state, the intracellular portion of a GPCR binds to a heterotrimeric G protein, and the G protein binds to a GDP molecule. During a cell signaling event, a typical GPCR binds an extracellular ligand that acts as an **agonist** to activate the GPCR. The activated GPCR then activates the coupled heterotrimeric G protein by catalyzing the **exchange of GDP for GTP**. The activated G protein can then activate additional downstream processes.

The passage states that the opsin proteins are GPCRs activated by the light-stimulated *cis-trans* isomerization of retinal rather than by the binding of an extracellular ligand. The passage further states that the **transducin** is the heterotrimeric G protein associated with opsins. The question states that only transducin and its *downstream* components have been purified; opsins, which act *upstream* of transducin, have *not* been purified. To activate transducin, GTP must be exchanged for GDP. In the absence of coupled opsin proteins, this can be accomplished by **high levels of GTP**, which can compete for the GDP/GTP binding site of transducin to activate it.

(Choice A) An electrical signal is the *consequence* of opsin and transducin activation. Application of electrical stimulation to an *in vitro* solution of purified transducin is unlikely to activate it.

(Choice C) In intact photoreceptor cells, light energy will activate opsins, which can then activate transducin. However, the question states that only transducin

and *downstream* effectors have been purified; opsins are *upstream* of transducin and are *not* available to activate it in the proposed experiment.

(Choice D) Although phosphorylation by **kinases** is a common **post-translational modification** that can activate various proteins, no evidence is given in the passage that transducin is activated by phosphorylation.

Educational objective:

G protein–coupled receptors are membrane proteins that activate their associated G protein upon stimulation. Activation of the associated G protein involves the exchange of GDP for GTP.

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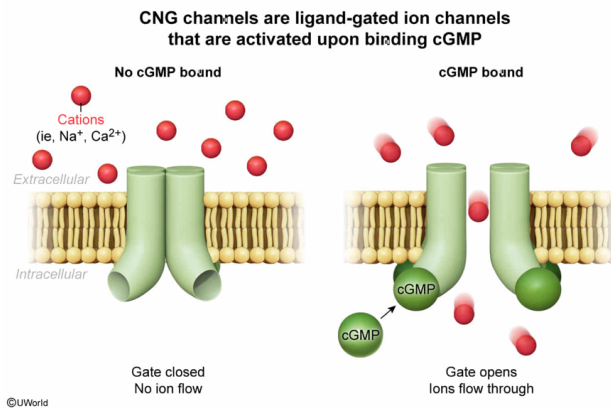
The CNG channels can be best described as:

- ✔ ☒ A. ligand-gated ion channels. [88%]
- ☐ B. mechanically gated ion channels. [2%]
- ☐ C. gap junction channels.
- ☐ D. voltage-gated ion channels. [9%]

Correct

87% answered correctly

Explanation:



The alteration of the **membrane potential** to send electrical signals is a common means for cells to communicate and/or amplify signals. To alter the membrane potential, cells can open **gated ion channels**. Gated ion channels are broadly classified based on the type of stimulus to which they respond.

- **Voltage-gated ion channels** respond to prior changes in the membrane potential. Voltage-gated channels have voltage sensors in their transmembrane regions (eg, α -helices rich in **positively charged amino acids**) that are electrostatically stabilized in a particular conformation at the cell's resting membrane potential. Nearby changes in membrane potential disrupt this stability, causing conformational changes that open the gate and allow the electrical signal to spread. The CNG channels in the passage are not described as responding to a change in membrane potential (**Choice D**).
- **Mechanically gated ion channels** respond to mechanical forces like pressure or tension. These forces alter the dimensions of the fluid lipid bilayer membrane, and therefore cause conformational changes that result in the opening of the gate. No such mechanical force is needed to open CNG channels (**Choice B**).
- **Ligand-gated ion channels** require a **ligand** to bind before the channel opens. Ligands that bind and activate receptors are also called **agonists**. Agonists function as allosteric activators of these channels because they bind to an extracellular or intracellular ligand binding site to cause a conformational change that affects the ion gate within the transmembrane domain of the channel. According to the passage, **cGMP** is an **intracellular ligand** that binds to CNG channels, **causing CNG channels**

to open.

(Choice C) Gap junctions are **cell-cell connections** that physically connect the cytosol of two cells. These junctions form pores that are large enough for small molecules and ions to pass through and electrically connect two cells together without the need for neurotransmitter release. In the passage, CNG channels allow *extracellular* sodium and calcium ions to enter, demonstrating that they do not connect the intracellular cytosols of two cells.

Educational objective:

Ion channel activity is crucial to modulating the membrane potential and allowing electrical cell signaling. Ligand-gated ion channels are ion channels that only open if they bind their appropriate ligand.

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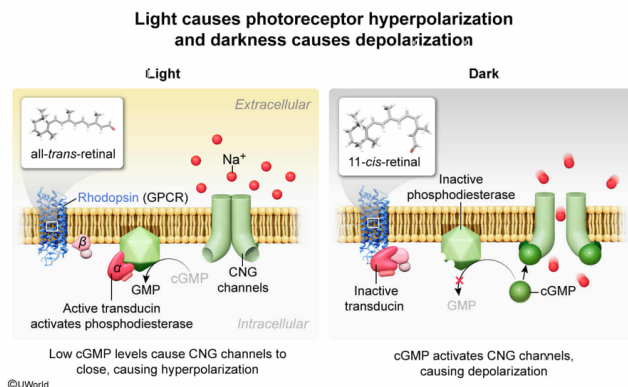
The information given in the passage suggests that rod and cone cells will respond to the sudden absence of light by:

- ✔ ☒ A. depolarizing. [28%]
- ☐ B. hyperpolarizing. [40%]
- ☐ C. generating an action potential. [10%]
- ☐ D. signaling to downstream cells without altering their own membrane potential. [21%]

Correct

28% answered correctly

Explanation:



The outcomes of biochemical pathways can be predicted by tracing the immediate effects of each step from the inciting stimulus to the outcome. The passage states that **stimulation by light** causes *cis-trans* isomerization of retinal, which **activates the GPCR** opsins. This **activates the G protein** transducin, which subsequently **activates** the downstream effector **cGMP phosphodiesterase**. Phosphodiesterases **degrade** cyclic nucleotides (eg, cGMP), meaning that cGMP can no longer bind to CNG channels. This lack of binding causes the channels to close. Therefore, light stimulation **prevents CNG channels from opening**.

When CNG channels are closed, cations cannot enter the cell. The resulting lack of incoming cytosolic positive charges, coupled with the continued outflow from the various pumps (eg, Na^+/K^+ pump), means that **light stimulation causes photoreceptor cells to hyperpolarize** by becoming more negative inside the cell than outside.

However, the question asks about the sudden **absence** of light. When light is removed, opsins and their associated transducins become *inactive*. Without transducins stimulating them, cGMP phosphodiesterase similarly becomes inactive, allowing cGMP to accumulate and bind to CNG channels, opening them. This increase in open CNG channels allows positively charged sodium and calcium ions to enter the cell, resulting in **depolarization** in the absence of light (ie, onset of darkness).

(Choice B) The removal of light causes *depolarization*, not hyperpolarization. Hyperpolarization is instead caused by the *presence* of light.

(Choice C) **Action potentials** are an **all-or-none response** generated by particular neuronal cells (eg, the ganglion cells) once the membrane potential

has crossed a threshold level. However, the passage states that photoreceptor cells show graded responses rather than an all-or-none response.

(Choice D) The passage states that light and dark stimulation alters ion channel activity through a biochemical pathway. The alteration of ion channel activity results in an alteration to the membrane potential of the cell receiving the signal.

Educational objective:

The outcome of a biochemical pathway can be predicted if the individual steps of the pathway are known. If the response to the presence of a stimulus is known, the response to the absence of that stimulus can be reasonably assumed to be the opposite.

Time spent:0 sec

Subject:
Biochemistry

QID:403745

System:
Amino Acids and Proteins

Protein isoelectric points (pI values) are a classical means of characterizing and distinguishing different proteins from each other and from their isoforms and posttranslationally modified variants. Owing to the importance of the pI value, bioinformatics studies were performed on the proteomes of various species to catalog the known pI value of each protein. When experimental pI values were not available, predictions were made based on the protein's translated mRNA sequence and theoretical estimates of side chain pK_a values (eg, using the measured pK_a values of free amino acid side chains).

Upon analyzing the data, researchers found that the proteomes of most species, including *Homo sapiens*, displayed mean and median pI values near physiological pH (ie, 7.4). However, the distribution was bimodal, with one mode near pI 6.0 and another mode near pI 8.25 (Figure 1). By contrast, proteins with pI values near 7.4 are underrepresented in most species' proteomes.

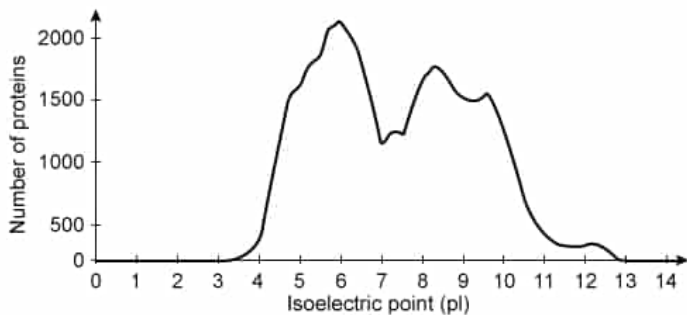


Figure 1 Distribution of pI values for proteins in the human proteome

Further analysis of the proteome showed that the protein pI value distribution varied with subcellular distribution. Cytosolic protein pI values tended to cluster in the acidic mode of the pI distribution, whereas the alkaline mode comprised a greater proportion of integral membrane proteins. Within organelles, the distributions remained bimodal but the mean pI value correlated with organellar pH. For example, in lysosomes (pH = 5), the mean pI value for proteins is near 5, but with most proteins having pI values either slightly above or slightly below 5.

A typical plasma membrane protein with a pI value of 8.0 is trafficked to

an endolysosome prior to its degradation. How will the membrane protein's net charge change as the endolysosome acidifies from pH 7.4 to pH 5? The protein's charge changes from:

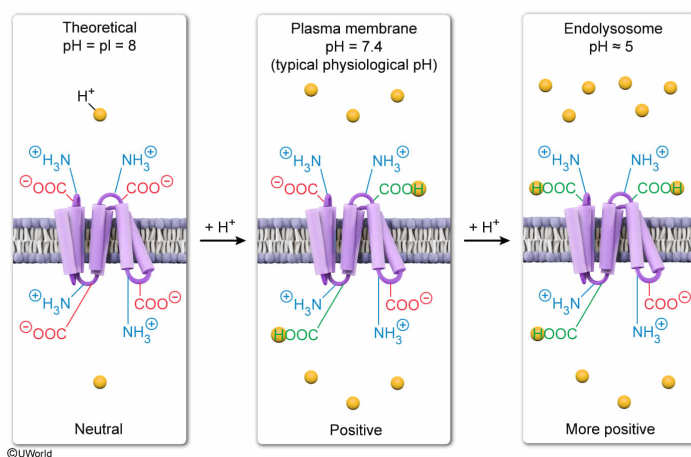
- ☐ A. positive to negative. [23%]
- ✓ ☒ B. positive to more positive. [45%]
- ☐ C. negative to positive. [22%]
- ☐ D. negative to more negative. [8%]

Correct

45% answered correctly

Explanation:

A protein's charge becomes more positive as the pH becomes more acidic



The **isoelectric point (pI)** of a protein is the pH value at which the protein has an overall **neutral** charge. Changing the pH of a solution changes not only the concentration of positively charged protons, but also the protonation state of any proteins within that solution. Consequently, the sign of the **net charge** of a protein (ie, whether it is positive or negative) can be determined by comparing the protein's pI value to the surrounding pH.

The question states that the membrane protein has a pI value of 8.0. Therefore, the protein would be neutral at a pH of 8.0. The cell, however, typically maintains a physiological pH of 7.4. This **pH is lower than the pI**, indicating that the pH around the plasma membrane is **more acidic** than the pI. As a result, the protein is partially **protonated** relative to its

neutral form, and therefore it is **positively charged** prior to entry into the endolysosome.

The protein is then trafficked to the endolysosome, where its environment becomes further **acidified**. Acidification involves the addition of positively charged protons. Some of these added protons attach to the protein, causing the protein to become **even more positively charged**.

(Choice A) Acidification (ie, the addition of positively charged protons) within the endolysosome will cause protein components within the endolysosomal lumen to become *more* positively charged, *not* negatively charged.

(Choices C and D) At the plasma membrane, the protein will experience a physiological pH of 7.4. This pH is *lower* (ie, more acidic, higher proton concentration) than the pI value of 8.0, which means that the protein is initially *positively charged* at the plasma membrane, *not* negatively charged.

Educational objective:

The isoelectric point (pI) of a protein is the pH value at which the protein has a net neutral charge. If the pH is lower than the pI, the protein is positively charged.

Time spent:0 sec

Subject:
Biochemistry

QID:403746

System:
Amino Acids and Proteins

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Upon analyzing the data, researchers found that the proteomes of most species, including *Homo sapiens*, displayed mean and median pI values near physiological pH (ie, 7.4). However, the distribution was bimodal, with one mode near pI 6.0 and another mode near pI 8.25 (Figure 1). By contrast, proteins with pI values near 7.4 are underrepresented in most species' proteomes.

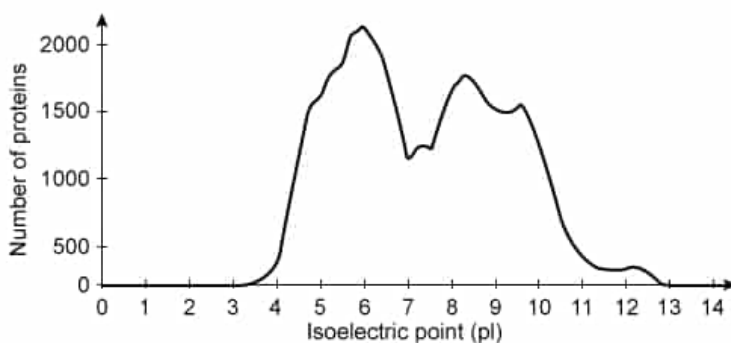


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pH. For example, in lysosomes (pH = 5), the mean pI value for proteins is near 5, but with most proteins having pI values either slightly above or slightly below 5.

Which of the following does NOT explain why the proteome pI distribution is bimodal?

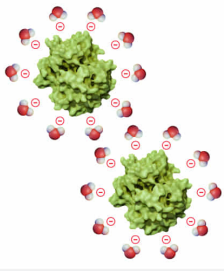
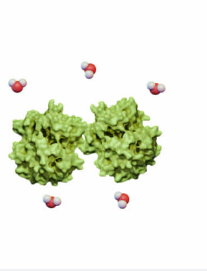
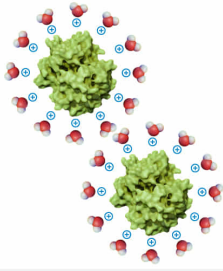
- ✓ ☒ A. Proteins are typically most active when pH exactly equals pI, so a bimodal distribution allows a different population of proteins to activate upon pH disturbance. [27%]
- ☐ B. Proteins are electrically neutral and therefore more prone to aggregation when pH = pI, resulting in an underrepresentation of pI values of 7.4. [36%]
- ☐ C. Most proteins are more soluble if their pI is either above or below, but not equal to, the surrounding pH. [18%]
- ☐ D. The pK_a distribution of the amino acid side chains is bimodal, leading to a bimodal distribution of pI values in the proteins they compose. [16%]

Correct

27% answered correctly

Explanation:

Protein pI values are related to their stability, solubility, and side chain pK_a values

pI = 6; pH = 7.4	pI = pH = 7.4	pI = 8; pH = 7.4
		
• Negative charge keeps protein soluble in water • Solvation shell keeps protein stable • Rich in low pK_a amino acids (eg, Glu, Asp) • Common (acidic mode) and active at pH 7.4	• Lack of charge minimizes solubility • Protein less stable (more prone to aggregation) • Rich in non-ionizable amino acids • Uncommon but still active at pH 7.4	• Positive charge keeps protein soluble in water • Solvation shell keeps protein stable • Rich in high pK_a amino acids (eg, Lys, Arg) • Common (alkaline mode) and active at pH 7.4

The **isoelectric point (pI)** of a protein equals the pH value at which the protein bears an overall neutral charge. For simple molecules, the pI is **estimated** based on the ionizable functional groups' pK_a values by taking the average of the **two adjacent** pK_a values surrounding the species with zero charge. Because the amino acid side chain **pK_a values** are clustered at either low values (roughly 4, 4, and 6 for Asp, Glu, and His, respectively) or high values (roughly 8, 10, 10.5, and 12 for Cys, Tyr, Lys, and Arg, respectively), the pI values of proteins derived from them also tend to cluster around low or high values (ie, the pI distribution is bimodal) (**Choice D**).

Tertiary and quaternary structure of proteins are driven and stabilized by several forces. Hydrophobic amino acids do not interact favorably with water and therefore are typically buried in the protein interior through **hydrophobic interactions**. Conversely, hydrophilic amino acids—especially **charged amino acids**—are commonly found at the protein **surface**. Charged amino acids play an important role in protein **solubility and stability**.

When pH equals pI, **ionizable surface residues** are **neutralized** either by protonation state or by interaction with another surface residue. This neutralization lessens the favorability of interactions with water, **decreasing solubility** and **promoting aggregation** (ie, the protein is less stable). To prevent this, most proteins do not have a pI of 7.4, but instead have pI values either above or below physiological pH (**Choices B and C**).

The question asks for the statement that is *unlikely* to explain why the proteome pI distribution is bimodal. The activity of *most* proteins is tuned to be **highest at their physiological pH**, not at their pI. Therefore, they would *not* be expected to be activated by disturbances to the ambient pH. Instead, optimal activity, which is tuned to physiological pH (ie, 7.4), is expected to be **unimodal**.

Educational objective:

The isoelectric point (pI) is derived from the pK_a values of the ionizable amino acids that make up the protein; when $pH = pI$, proteins have an overall neutral charge and are at their least soluble and least stable pH. The optimal pH for protein activity is independent of pI and instead reflects its physiological environment.

Time spent:0 sec**Subject:**
Biochemistry**QID:**403747**System:**
Amino Acids and Proteins

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Upon analyzing the data, researchers found that the proteomes of most species, including *Homo sapiens*, displayed mean and median pI values near physiological pH (ie, 7.4). However, the distribution was bimodal, with one mode near pI 6.0 and another mode near pI 8.25 (Figure 1). By contrast, proteins with pI values near 7.4 are underrepresented in most species' proteomes.

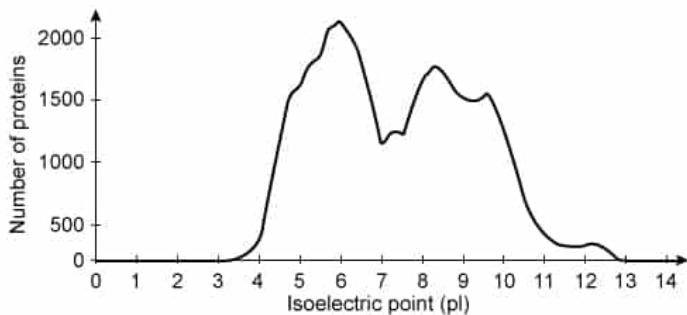


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A researcher seeks to study proteins found in the relatively high-pH

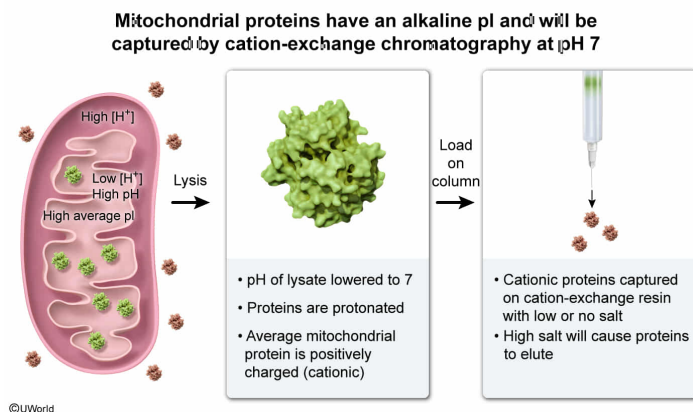
environment of the mitochondrial matrix. Based on the information in the passage, which of the following procedures will be most successful at causing mitochondrial matrix proteins to bind to an ion-exchange chromatography column?

- ☐ A. Placing the proteins in high-salt buffer at pH 7 and then running it through an anion-exchange column. [22%]
- ✓ ☒ B. Placing the proteins in low-salt buffer at pH 7 and then running it through a cation-exchange column. [31%]
- ☐ C. Placing the proteins in low-salt buffer at pH 7 and then running it through an anion-exchange column. [24%]
- ☐ D. Placing the proteins in high-salt buffer at pH 7 and then running it through a cation-exchange column. [21%]

Correct

31% answered correctly

Explanation:



Ion-exchange chromatography is a protein purification technique that separates proteins by their overall net charge. **Cationic** (positively charged) proteins bind to **cation-exchange chromatography** columns, whereas anionic (negatively charged) proteins bind to anion-exchange chromatography columns. The proteins must be **loaded** onto the column in **low-salt conditions** to allow them to bind. After cationic proteins are bound to the column, they can then be *removed* (ie, eluted) by increasing the salt concentration. This is because **ions** from the salt compound bind to the column and **exchange** places with the proteins. The proteins are collected as they exit the column, resulting in a purified protein sample.

The question asks how best to bind **mitochondrial matrix** proteins. Mitochondria are the organelles in eukaryotic cells where the **electron transport chain (ETC)**, **oxidative phosphorylation**, and **ATP synthesis** occur. The ETC pumps protons to the intermembrane space, and oxidative phosphorylation occurs as protons move **down** their **electrochemical gradient** back **into** the mitochondrial matrix to promote ATP synthesis. This proton gradient is constantly maintained, and the **mitochondrial matrix** therefore maintains a low proton concentration and a relatively **high pH**.

The passage indicates that the mean (ie, average) isoelectric point (pI) of proteins tends to be near the pH of their subcellular compartment. Therefore, the **average pI value** of mitochondrial matrix proteins is likely to be **alkaline (above 7)**. Consequently, if the pH of the buffer is 7, most matrix proteins will be **protonated, positively charged, and cationic**

when placed in this buffer. Therefore, they will bind a **cation-exchange column** with **low salt concentration**. (Note: Due to the bimodal distribution of pI values, not *all* matrix proteins will be captured; however, this condition will be the *most* successful at maximizing their capture.)

(Choices A and D) Salt is used to *elute* and remove proteins from ion-exchange columns. Therefore, proteins are unlikely to bind a column in a high-salt buffer regardless of charge.

(Choice C) Most mitochondrial proteins will be positively charged and therefore will be captured on a *cation*-exchange column, not an anion-exchange column.

Educational objective:

The mitochondrial matrix is a high-pH environment and has proteins with high isoelectric point (pI) values. High-pI proteins will be positively charged at pH 7, allowing them to bind to a cation-exchange chromatography column at low salt concentrations.

Time spent: 0 sec

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Biochemistry

QID: 403748

System:
Amino Acids and Proteins

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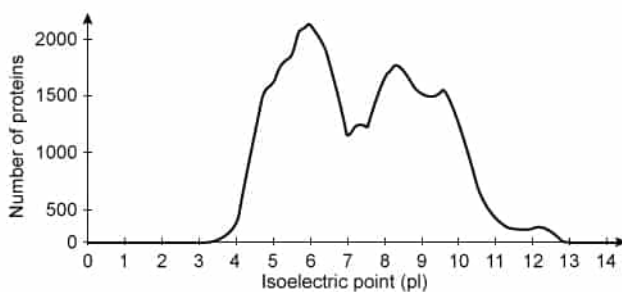


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Researchers purified protein X, experimentally measured its pI value through isoelectric focusing (IEF), and found that it differed significantly from the theoretical pI calculated as described in the passage. The theoretical pI value may have differed from the experimental pI if the theoretical calculation did not consider:

- I. posttranslational modifications on protein X.

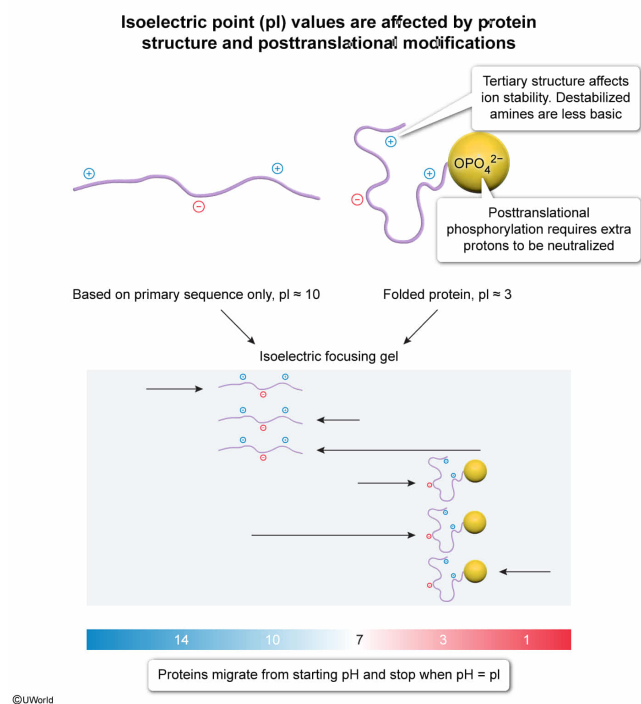
- II. the tertiary and quaternary structure of protein X.
III. the pH in the subcellular localization of protein X.

- ☐ A. I only [10%]
✓ ☒ B. I and II only [29%]
☐ C. II and III only [14%]
☐ D. I and III only [45%]

Correct

30% answered correctly

Explanation:



The **isoelectric point (pI)** of a protein is the pH value at which the protein bears an overall neutral charge. The pI is experimentally measured through a technique called **isoelectric focusing (IEF)**. In IEF, proteins are loaded onto a gel containing an immobilized pH gradient. An electric field is applied, and ionized proteins migrate across the pH gradient based on their charge. When $pH = pI$, the protein becomes electrically neutral, is no longer affected by the electric field, and stops migrating, allowing for experimental determination of the pI value.

A protein's pI value depends on the pK_a values of its **ionizable functional groups**. Ionizable functional groups may be located on the protein's **amino acids** or on a **posttranslational modification** (eg, phosphorylation). The addition of any **ionizable posttranslational modification** is likely to **alter the pI of a protein** relative to the theoretical pI calculated on the unmodified protein (**Number I**).

Further differences between theoretical and experimental pI values may also arise if the theoretical pI value was calculated using the pK_a values of free amino acids. When incorporated into a peptide, the ionizable functional groups will be affected by electrostatic interactions with nearby groups. A fully **folded protein** can position ionizable residues very close together, where they can be **stabilized in a salt bridge** or **destabilized via charge–charge repulsion**. These forces alter the equilibrium of proton dissociation, thereby **altering the K_a and pK_a**

values for each residue, resulting in an **altered pI of a protein** relative to the theoretical pI calculated using free amino acids (**Number II**).

(Number III) Neither the pH of the subcellular location nor the protein's starting pH value when entering an IEF gel will affect its final experimental pI determination. A protein undergoing IEF will migrate toward its pI **regardless of its starting pH**. Therefore, the pH of the protein's subcellular localization is unlikely to cause a difference between the theoretical and experimental pI values.

Educational objective:

The isoelectric point (pI) is calculated based on the pK_a values of the ionizable functional groups on a protein and can therefore be affected by posttranslational modifications (which can add or remove functional groups) or by protein structure (which alters the pK_a values). The pI is measured by isoelectric focusing gels, which focus proteins toward their pI regardless of their starting pH.

Time spent:0 sec

Subject:
Biochemistry

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System:
Amino Acids and Proteins